

Homework 5

1) group $n \geq 4$ ppl comparing birthdays
↳ assume birthdays independent & not Feb 29

a) I_{ij} → indicator of i & j ($i < j$) having same birthday

I_{12} and I_{34} → yes, independent
no information provided
conditioning on either:
e.g. $P(I_{12}=1 | I_{34}=1) = P(I_{12})$

I_{12} and I_{13} → independent for some reason

I_{ij} 's are independent

correction: $P(I_{ij}=1 | I_{ik}=1 \& I_{jk}=1) \neq P(I_{ij})$
↳ generally independent but NOT always
pairwise

b) Poisson is applicable b/c
↳ large # of events $\binom{n}{2}$ pairs
↳ small prob of each event
↳ events are largely independent

$$n = 23$$

$$E(\# \text{ of matches}) = \binom{23}{2} \frac{1}{365} = \frac{253}{365} = \lambda$$

$$P(X \geq 1) = 1 - P(X=0) = 1 - e^{-\lambda} \approx 0.500$$

c) how many people needed so that 50% chance
a birthday match OR 1 day apart

$$E(\# \text{ of match } " \pm 1 ") = \binom{n}{2} \frac{3}{365}$$

need this to be $> \frac{253}{365}$

$$n = 13 \rightarrow E(X) = \frac{234}{365}$$

$$n = 14 \rightarrow E(X) = \frac{273}{365}$$

↑

2) T PDF: $\frac{1}{5} e^{-t/5}$ t in unit years

c.v. $X = 0$ posted in first day
 1 " " 2nd "
 $\vdots$ $\vdots$

find PMF of X

$$\begin{aligned} P(X=x) &= P\left(\frac{x}{365} \leq T < \frac{x+1}{365}\right) \\ &\uparrow \\ &\text{day posted} \\ &= \text{CDF}\left(\frac{x+1}{365}\right) - \text{CDF}\left(\frac{x}{365}\right) \\ &= (1 - e^{-\frac{1}{5} \cdot \frac{x+1}{365}}) - (1 - e^{-\frac{1}{5} \cdot \frac{x}{365}}) \\ &= e^{-\frac{x}{1825}} - e^{-\frac{x+1}{1825}} \\ &= e^{-\frac{x}{1825}} (1 - e^{-\frac{1}{1825}}) \end{aligned}$$

$X \sim \text{Geom}(1 - e^{-\frac{1}{1825}})$ $\uparrow$ # of failures $\uparrow$ until first success

3) $U \sim \text{Unif}(-1, 1)$

$$a) E(U) = \frac{1 + (-1)}{2} = 0$$

$$\text{Var}(U) = E(U^2) - (E(U))^2$$

$$= \int_{-1}^1 x^2 \frac{1}{2} dx \quad \text{by LOTUS}$$

$$= \frac{1}{6} x^3 \Big|_{-1}^1 = \frac{1}{3}$$

$$E(U^4) = \int_{-1}^1 x^4 \frac{1}{2} dx = \frac{1}{10} x^5 \Big|_{-1}^1 = \frac{1}{5}$$

b) CDF $F_X(x)$ $P(X \leq x) = P(-\sqrt{x} \leq U \leq \sqrt{x})$

$$= \frac{\sqrt{x} - (-\sqrt{x})}{2} = \sqrt{x}$$

PDF $F'(x) = \frac{1}{2} x^{-1/2} \rightarrow$ not a constant
 U^2 is not unif. dist.

4) F is CDF that's continuous & strictly inc.
 $F^{-1} \rightarrow$ 'quantile fn'

Universality Thm.

Let r.v. $X = F^{-1}(U)$, then $X \sim F$

given

$$\begin{aligned} \mu = E(X) &= E(F^{-1}(U)) \quad \text{by universality} \\ &= \int_{-1}^1 F^{-1}(u) F'(x) dx \\ &= \int_{-1}^1 F^{-1}(F(x)) F'(x) dx \\ &= \int_{-1}^1 x F'(x) dx \end{aligned}$$

$$\begin{aligned} \text{find } \int_0^1 F^{-1}(u) du &= \int_0^1 F^{-1}(F(x)) du \\ &= \int_0^1 x du = \int_0^1 x F'(x) dx = \mu \end{aligned}$$

$$\begin{aligned} X &= F^{-1}(U) \\ F(X) &= U \\ F'(x) dx &= du \end{aligned}$$


5) $Z \sim N(0,1)$

z device observes Z

but gives reading 0 if $z \leq 0$

assume $X = Z I_{Z>0}$

$I_{Z>0}$ is indicator of $z > 0$

$$E(X) = \int_{-\infty}^{\infty} I_{z>0} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= \int_0^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= -\frac{1}{\sqrt{2\pi}} e^{-z^2/2} \Big|_0^{\infty} = -\frac{1}{\sqrt{2\pi}} (0 - 1) = \frac{1}{\sqrt{2\pi}}$$

$$\text{Var}(X) = E(X^2) - (EX)^2 = \frac{1}{2} - \frac{1}{2\pi}$$

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$$\int_{-\infty}^{\infty} I_{z>0} z^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= \int_0^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{1}{2}$$

following same
integration by
parts method as
seen in lecture